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18 March 1965
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12 March 1965
(See 6.4)

MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, SYNTHETIC FIBER REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for synthetic fiber reinforced rubber hose for use with reusable end fittings for low, medium and high pressure service in water and oil systems.

1.2 Classification. Hose and end fittings shall be of the types and sizes specified in the individual specification sheets.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-775 - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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MILITARY (Continued)

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Servive.
MIL-H-5606 - Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-L-17331 - Lubricating Oil and Gear, Moderate Service.
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).
MIL-H-24136/1 - Hose, Synthetic Rubber, Synthetic Fiber Reinforced, and End Fittings, Reusable, For Flexible Hose Assemblies (Low Pressure, Sizes -80 through -192) and Adapters.
MIL-H-24136/2 - Hose, Synthetic Rubber, Synthetic Fiber Reinforced, and End Fittings, Reusable, For Flexible Hose Assemblies (High Pressure Sizes -4 through -32).
MIL-H-24136/3 - Hose, Synthetic Rubber, Synthetic Fiber Reinforced, and End Fittings, Reusable, For Flexible Hose Assemblies (Low Pressure Sizes -4 through -32).
MIL-H-24136/4 - Hose, Synthetic Rubber, Synthetic Fiber Reinforced, and End Fittings, Reusable, For Flexible Hose Assemblies (Low Pressure Sizes -40, -48 and -64) and Adapters.
MIL-B-24480 - Bronze, Nickel-Aluminum Castings, For Seawater Service.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards For Federal Services.
FED-STD-162 - Hose, Rubber, Visual Inspection Guide for.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-177 - Rubber Products, Terms for Visible Defects of.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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ASTM

D 380 - Rubber Hose. (DoD adopted)
D 413 - Rubber Property-Adhesion to Flexible Substrate, Test Method for. (DoD adopted)
D 1141 - Substitute Ocean Water. (DoD adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The hose and end fittings furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bid (see 4.3 and 6.3).

3.2 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.3 Material.

3.3.1 Hose. Hose shall consist of a synthetic rubber tube, reinforcement, and a synthetic rubber cover.

3.3.1.1 Rubber. The rubber compounds used shall be a synthetic rubber as specified below:

(a) Compounds using polymerized chloroprene as the basic material.
(b) Compounds using copolymer product of butediene as the basic material.

3.3.1.2 Reinforcement. The reinforcement shall consist of high quality polyester or aramid fibers which have been processed and treated to obtain high strength and low creep properties.

3.3.1.3 Frictioned fabric layer. The frictioned fabric layer, if used, between the tube and reinforcement shall be a cotton or synthetic fiber of good quality, or a combination of both.

3.3.2 Fittings. Fittings shall be of a material as specified on the applicable specification sheet.

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3.3.3 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3.4 Flanges. Material for flanges shall be nickel-aluminum-bronze in accordance with MIL-B-24480.

3.4 Design and construction.

3.4.1 Hose. The hose shall consist of a synthetic rubber tube, polyester or aramid fiber reinforcement, and a synthetic rubber cover. The hose shall be constructed to meet the requirements of this specification and to retain qualified end fittings without slippage or leakage when tested as specified herein.

3.4.2 Tube. Tube shall be fabricated from synthetic rubber and shall have a smooth bore. Tube shall be free from pitting, cuts, dirt, foreign material, mandrel lubricants or other defects that would affect the performance of the hose.

3.4.3 Reinforcement. The reinforcement shall consist of synthetic polyester or aramid fiber evenly and firmly braided or spirally wrapped in such a manner as to meet the requirements of this specification. There shall be no broken reinforcing fibers nor shall any reinforcing fibers be omitted in any length of hose. A frictioned fabric layer may be applied over the reinforcement.

3.4.4 Cover. Cover shall be made of synthetic rubber compounded to give maximum resistance to oxidation, weathering, abrasion, and light aging, and to meet the requirements of this specification.

3.4.5 Concentricity. Concentricity, based on full indicator reading between the inside bore of the hose and the outer surface of the hose, shall not exceed the following values: For size 1/4 $\pm$ 0.030 inch; for sizes over 1/4-inch, up to and including 2 $\pm$ 0.050 inches; for sizes over 2 $\pm$ 0.0625 inches.

3.4.6 Dimensions and pressures. Hose dimensions and pressures shall be in accordance with the applicable specification sheet.

3.4.7 Temperature range. Hose shall comply with specified requirements through a temperature range of minus 40 to 200°F for all applications.

3.4.8 Lengths. Hose furnished and shipped in straight lengths shall be not less than 12 feet in length. Hose length shall be the actual length measured along the cover.

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3.4.9 End fittings and flanges. End fittings and flanges shall be the types and sizes specified on figures to the applicable specification sheet and shall be of the reusable design. Hose end of the fitting may be of a solid socket, ring retained segmented socket, or bolt retained segmented socket or any other design capable of meeting all the requirements of this specification. Hose end of the fitting shall be readily removable and reattachable to a new section of hose by the use of commonly available tools which are normally carried aboard ship. Flanges shall be in accordance with figure 1, as applicable.

3.4.10 Weights and dimensions (fittings). Weight of the end fitting shall be kept to a minimum and the dimensions of the fitting shall not exceed the envelope limits shown on figures to the applicable specification sheet.

3.4.11 Finish. Fittings shall be finished smooth and free from nicks, scratches, burrs and sharp edges.

3.4.12 Tool and die marks (nickel copper alloy bar stock only). The presence of tool marks on machined parts is acceptable provided such marks do not exceed 3 percent of the wall thickness or reduce the wall thickness to an unacceptable level. Draw marks upon the external hexagonal surfaces, such as the outside diameter (o.d.) of solid sockets, which are normally unmachined surfaces, are acceptable if they do not exceed 3 percent of the wall thickness.

3.4.13 Welding. Welding, if used, shall be in accordance with MIL-STD-278. Brazing of fitting parts shall not be permitted. If welding is used in the fabrication of any of the types of fittings covered by this specification, welding procedures used by the contractor for the materials specified herein, shall be submitted to NAVSEA for information. Procedures shall be submitted prior to the start of qualification testing.

3.4.13.1 Weld repair. Weld repair shall be in accordance with MIL-STD-278.

3.4.14 Mating. Where two parts of the same material are mated together, provision shall be made to prevent galling and seizing of the mated surfaces. A solid film lubricant, such as molybdenum disulfide, or two classes of treatments of the same materials, or a combination of any of these methods may be used to prevent galling and seizing.

3.5 Physical requirements.

3.5.1 Proof pressure. Hoses shall show no evidence of leakage, rupture, or deformation when subjected to the proof pressure test specified in 4.6.1.

3.5.2 Stability (dimensional change). When tested in accordance with 4.6.2, hoses shall not change in length more than plus 2 percent or minus 4 percent.

3.5.3 Aging. The hose assembly shall show no evidence of aging, such as cracking, separation of plies, or any deformation in the form of bulging or collapsing when subjected to tests specified in 4.6.3.

3.5.4 Hydraulic fluid circulation. There shall be no evidence of leakage through the hose or hose fitting when subjected to the hydraulic fluid test specified in 4.6.4.

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3.5.5 Hydraulic impulse. Hose shall show no evidence of failure or leakage when subjected to the hydraulic impulse test specified in 4.6.5.

3.5.6 Burst pressure. Hoses shall show no indication of failure or leakage at less than the specified burst pressure when subjected to the burst test specified in 4.6.6.

3.5.7 Vacuum. There shall be no evidence of ply separation or other degradation of the hose when vacuum tested as specified in 4.6.7.

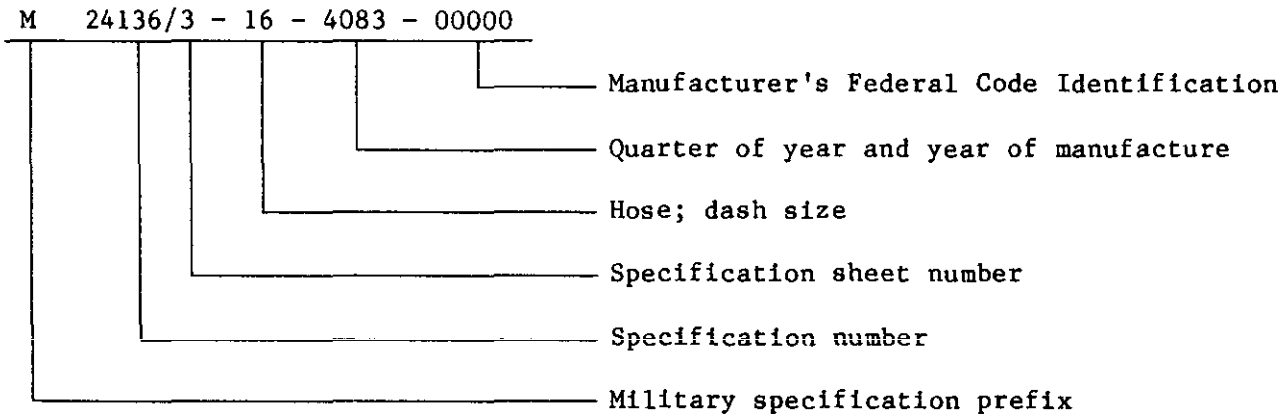
3.5.7.1 Internal support devices. When required for test purposes, internal support devices shall be fitted to the hose to prevent collapse under vacuum service. Support devices of the coil type shall be a flat wound spring wire strip or bonded rubber/ring support cylinder. Coil type support devices for hose sizes -64 and larger shall be fitted with spacer strips to keep coil properly positioned in the hose.

3.5.8 Cold temperature flexibility. Hose shall show no evidence of cracking or failure when subjected to the test specified in 4.6.8. Cold temperature flexing of hoses larger than size -32 will be waived in lieu of a certified affidavit that rubber compounds meet the requirements of ASTM D 380.

3.5.9 Adhesion between piles. The adhesion between the tube, cover and reinforcement shall be not less than 15 pounds per square inch (lb/in^2) when tested as specified in 4.6.9.

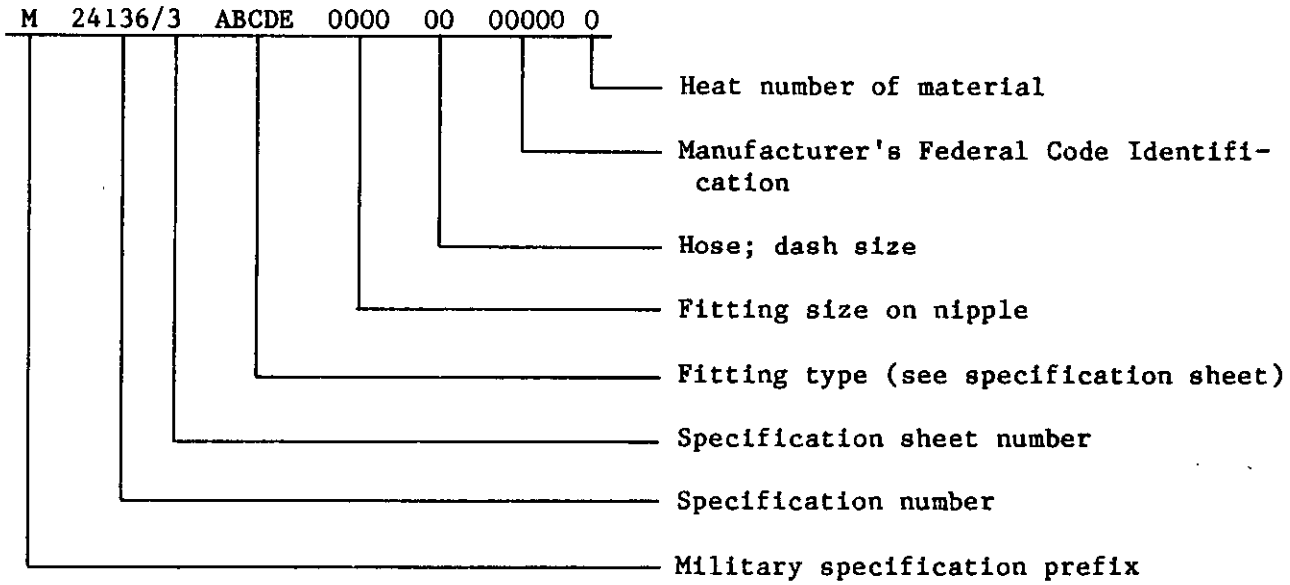
3.6 Identification marking.

3.6.1 Hose. Hose shall have a permanent and visible embossed or inlaid layline along its entire length. The numerals and lettering shall be not less than 3/16-inch high, and each set of markings shall be repeated at intervals not exceeding 20 inches. The layline shall contain, but not be limited to, the following:



3.6.2 End fittings. Each end fitting part shall have permanent and visible stamped or embossed identification. The numerals and lettering shall be as large as possible, but not less than 1/8-inch high (see figure 1). The identification shall contain the following:

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3.6.3 Instructions. Instructions for installing the fittings on to the hose shall be packed with each fitting and shall contain, but not be limited to, the following:

(a) Steps and lists of tools required to properly attach the fitting to the hose. Instructions shall be complete, clear and concise.
(b) Maintenance or assembly precautions required.

The instruction tag shall be a permanent type that will not become illegible or be damaged during packaging or storage. The manufacturer shall furnish a copy of the instructions to NAVSEA for review as part of his qualification test report.

3.7 Threads. Where screw threads are required, they shall be in accordance with FED-STD-H28.

3.8 Workmanship. Hoses and fittings shall be uniform in quality and in material. Castings shall be free from patching, misalinement resulting from shifted coring, warping, porosity or other defects. Sprues shall be removed, and the castings shall be free from sand, dirt, scale, and other extraneous materials. Forgings shall be free from patching, warping or other defects and free from dirt, scale, and other extraneous materials. Machined parts shall be manufactured to tolerances and dimensions specified herein. Hose and fitting shall be free from workmanship deficiencies that could impair the function or serviceability of the hose and fitting in its intended use.

3.9 Age of manufacture (hose). Hose shall be manufactured not more than 8 calendar quarters (2 years) prior to date of delivery.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with the contract or order (see 6.2.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command. Qualification inspection shall consist of the examination of 4.5 and the tests of 4.6.1 through 4.6.9. Required hose assemblies and sequence of tests shall be as specified on the applicable specification sheet. If end fittings only are being qualified, the stability test (see 4.6.2), aging test (see 4.6.3), and cold temperature flexibility test (see 4.6.8) need not be conducted.

4.3.1 Sampling for qualification. Sampling of hose, end fittings, or hose assemblies and the sequence of tests shall be in accordance with the applicable specification sheet.

4.3.2 Fitting types required. If the hose end of the fitting is identical for more than one type of fitting, only a representative type of fitting will require qualification testing. Qualification may be granted to all those types which the sample represents, subject to the results of the tests.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination of 4.5 and tests specified in 4.6.1, 4.6.2 and 4.6.6.

4.4.1 Lots.

4.4.1.1 Hose. A lot shall consist of hose at the same size and type offered at one time for purposes of quality conformance inspection. At the discretion of the contractor and the Government inspector, hose in excess of contractual requirements may be offered for quality conformance inspection provided the hose is identified with a lot number in the layline, in addition to other identification requirements.

4.4.1.2 Fittings. A lot shall consist of fittings of the same type and size offered at one time for the purpose of quality conformance inspection.

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4.4.1.3 Hose assemblies. A lot shall consist of all hose assemblies made of the same type and size hose offered at one time for purposes of quality conformance inspection.

4.4.2 Sampling for visual and dimensional examination.

4.4.2.1 Hose. A random sample of hose shall be selected from each lot (see 4.4.1.1) in accordance with MIL-STD-105 for normal level of examination using a major defective acceptable quality level (AQL) value of 1.0 percent and a 2.5 percent defective for minor defects. Examination shall be in accordance with 4.5 and table I.

4.4.2.2 Fittings. A random sample of fittings shall be selected from each lot (see 4.4.1.2) in accordance with MIL-STD-105 for normal level of examination using a major defective acceptable quality level (AQL) value of 1.0 percent and a 2.5 percent defective for minor defects. Examination shall be in accordance with 4.5 and table I.

TABLE I. Classification of defects.

Category	Defects
Critical	
1	Hose does not conform to dimensions and tolerances specified
2	Reinforcement not as specified
3	Hose size not as specified
4	Length not as specified
5	Marking missing or not as specified
6	Missing braid or spiral ply
7	Crack or hole in tube
8	Fittings or hose assemblies not as specified
Major	
101	Tube not as specified
102	Broken ply
Minor	
201	Depressed area of hose (exceeds outside diameter minimum tolerance)
202	Depressed area of hose tube (exceeds inside diameter maximum tolerance)
203	Presence of foreign material

4.4.3 Sampling for tests.

4.4.3.1 Hose. A random sample of hose, in 18-inch lengths, shall be selected from each lot in accordance with MIL-STD-105 and subjected to the length change and burst tests specified in 4.6.2 and 4.6.6. The inspection level shall be normal and the AQL 1.5 percent.

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4.4.3.2 Fittings. A random sample of fittings shall be selected from each lot in accordance with MIL-STD-105 and subjected to the burst test of 4.6.6. The inspection level shall be normal and the AQL 1.5 percent. In performing these tests, the fittings shall be fitted onto hose lengths (not less than 15 inches in length) which have been previously qualified.

4.4.3.3 Hose assemblies. Hose assemblies shall be fabricated from hose and fittings which have been previously examined in accordance with 4.4.2.1 and 4.2.2.2 and tested as specified in 4.4.3.1 and 4.4.3.2. Hose assemblies shall be examined dimensionally to the requirements of the contract and proof-tested as specified in 4.6.1.

4.5 Examination.

4.5.1 Hose. Each of the hose samples selected in accordance with 4.4.2.1 shall be visually and dimensionally examined to determine conformance with the applicable specification sheet and any other requirements of this specification not involving tests. Sample hose containing one or more visual or dimensional defects shall be rejected, and if the number of defective hose samples exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Evaluation of defects shall be in accordance with FED-STD-162. Forms for visible defects shall be as defined in MIL-STD-177.

4.5.2 End fittings. Each sample fitting selected in accordance with 4.4.3.2 shall be visually and dimensionally examined to determine conformance with 3.4.9. Any sample fitting containing one or more visual or dimensional defects shall be rejected, and if the number of defective fittings exceed the acceptance number for that sample, the lot represented by the sampling shall be rejected.

4.6 Tests.

4.6.1 Proof pressure. The proof pressure test of the hose shall be conducted in accordance with ASTM D 380 except that the samples shall be in accordance with 4.3.1 and 4.4.1 and the pressure shall be increased at a rate not less than 15,000 lb/in^2 per minute nor more than 25,000 lb/in^2 per minute. The proof pressure shall be 200 percent of the maximum working pressure specified in the applicable specification sheet, and shall be held for not less than 1 minute nor more than 5 minutes. Hoses which pass during this test are acceptable for use. Evidence of leakage, rupture, or deformation of the hose shall constitute failure of this test.

4.6.2 Stability (dimensional change). After successful completion of the proof test, the hose samples shall be measured as specified in ASTM D 380 except that they shall be marked at least 1/2 inch from each end fitting and the distance measured between the marks. The maximum working pressure specified in the applicable specification sheet shall then be applied and the distance between the marks measured at that pressure. Nonconformance to 3.5.2 shall constitute failure of this test.

4.6.3 Aging of samples. Where required, the assembly shall be completely filled and pressurized to working pressure with the hydraulic fluid in accordance with MIL-L-2104, MIL-H-5606 or MIL-L-17331, and held at this pressure while

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immersed in a nonpressurized, closed-type container or a reflux-type condenser to prevent distillation of the volatile matter in the fluid. Each assembly shall be aged by soaking it in the fluid specified on the applicable specification sheet. Each assembly shall be aged by being subjected to a temperature not less than 158°F nor greater than 160°F first in air, then immersed in synthetic sea water conforming to ASTM D 1141, for the times specified below:

(a) 76 hours plus or minus 1/4 hour in air.
(b) 46 hours plus or minus 1/4 hour in synthetic sea water.

Provision shall be made to bleed off the excess pressure caused by thermal expansion of the pressurized fluid. Nonconformance to 3.5.3 shall constitute failure of this test.

4.6.4 Hydraulic fluid circulation. Sample hose assemblies shall be filled with hydraulic fluid in accordance with MIL-L-2104, MIL-H-5606 or MIL-L-17331. Sample hose assemblies shall then be pressurized from 75 to 100 lb/in^2. The fluid temperature shall be raised from ambient temperature to 180 $\pm$ 10°F and the fluid circulated through the hose assembly at a flow rate of not less than 3 gallons per minute. The ambient temperature shall be maintained at 75 $\pm$ 15°F. Every 24 hours the pressure shall be increased to maximum working pressure listed in the applicable specification sheet and maintained for not less than 5 minutes. The flow shall be continued for not less than 200 hours. At the completion of this test, the hose assembly shall be pressurized to maximum working pressure specified in the applicable specifications sheet, with the pressure held for not less than 5 minutes. Nonconformance to 3.5.4 shall constitute failure of this test.

4.6.5 Impulse. Hose assemblies shall be subjected to impulse testing to the peak pressure specified on the applicable specification sheet. Hose assemblies shall be subjected to impulse rate from 30 to 80 cycles per minute, with the hydraulic fluid at a temperature of not less than 120°F nor greater than 130°F for a total of 150,000 cycles. Hose assemblies in sizes -4 through -12 shall be bent through 180 degrees to the minimum bend radius specified in the applicable specification sheet during the impulse test, with both ends rigidly supported. Hose assemblies larger than size -12 shall be dynamically impulsed straight with one end capped and free from support. At the beginning of each pressure cycle, the peak pressure specified on the applicable specification sheet shall be attained. The applicable working (service) pressure shall be attained before leveling off (see figure 2 or 3 as applicable).

4.6.5.1 Retest and averaging impulse. In the event of a hose failure during impulse testing, an additional hose assembly may be tested and the results of all assemblies of the same size tested shall be averaged. The conditions for testing an additional hose assembly are as follows:

(a) The failed hose shall have reached not less than 95 percent of the required number of impulse cycles.
(b) For the purpose of averaging the number of cycles, the maximum number of cycles that can be used from any single tested hose assembly is 110 percent of the required number of cycles, or the actual number of cycles at the time of failure if failure occurs between 100 and 110 percent of the required number of cycles.

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Nonconformance to 3.5.5 shall constitute failure of this test.

4.6.6 Burst. Sample hose assemblies shall be subjected to increasing internal cold hydraulic pressure at a rate not less than 15,000 lb/in^2 per minute, nor more than 25,000 lb/in^2 per minute until failure occurs. The pressure at which failure occurs shall be recorded. Hose assemblies utilized during this test are not acceptable for use. Nonconformance to 3.5.6 shall constitute failure of this test.

4.6.7 Vacuum. The specified hose shall be additionally tested under a vacuum not less than 29 inches of mercury for not less than 30 seconds. Nonconformance to 3.5.7 shall constitute failure of this test. If the size hose dictates support, internal support shall be used during the vacuum test.

4.6.8 Cold temperature flexibility. The sample hose assemblies shall be cold-temperature-flexibility tested while filled with fluid as specified in the applicable specification sheet and enclosed in an environmental chamber of minus 40 $\pm$ 2°F for not less than 24 hours. At the end of this period and while at this temperature, the hose assemblies sized -4 to -16 shall be flexed through not less than 180 degrees, to the minimum bend radius specified in the applicable specification sheet within 12 seconds. For size -16 and larger, the flexing shall be through not less than 90 degrees to the minimum bend radius specified in the applicable specification sheet within 12 seconds. Nonconformance to 3.5.8 shall constitute failure of this test. For size -32 and larger, a certified affidavit that hose meets requirements of ASTM D 380 is acceptable in lieu of performing the above test.

4.6.9 Adhesion. Adhesion of the tube and cover to the reinforcement, and of plies of reinforcement to each other shall be determined in accordance with ASTM D 413 except that the required minimum weight shall be supported by the test specimen for 3 minutes without causing a separation of more than 1 inch. Three specimens of each interface shall be tested. Nonconformance with 3.5.9 shall constitute failure of this test.

4.7 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5).

5.1 Preservation-packaging, packing and marking. Hose, hose assemblies and end fittings shall be preserved-packaged level A or C, packed level A, B or C as specified (see 6.2.1.1 and 6.2.1.2) and marked in accordance with MIL-P-775.

5.2 Talc/tacum. Talc/tacum used in the packaging process of items shall be free of asbestos and asbestiform-like materials.

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6. NOTES

6.1 Intended use. Hoses and end fittings are intended for use in piping systems as flexible connections to resiliently mounted (sound mounted) equipment but may be used for other applications which do not exceed the pressure rating of the hose selected.

6.2 Ordering data.

6.2.1 Acquisition requirements.

6.2.1.1 Bulk hose and end fittings. For bulk hose and end fittings, acquisition documents should specify the following:

(a) Title, number and date of this specification and applicable specification sheet.
(b) Hose type and size (see 1.2).
(c) Type of end fittings (see 1.2).
(d) End fitting size (pipe end and hose end) (see 3.4.9).
(e) Length of hose required (see 3.4.8).
(f) Inspection system required (see 4.1.1).
(g) Level preservation-packaging, and packing (see 5.1).
(h) Total quantity for each size.

6.2.1.2 Hose assemblies. For hose assemblies, the acquisition documents should specify the following:

(a) Title, number and date of this specification and applicable specification sheet.
(b) Type of end fitting (see 1.2).
(c) Hose type and size (see 1.2).
(d) End fitting size (pipe end and hose end) (see 3.4.9).
(e) Inspection system required (see 4.1.1).
(f) Type of configuration.
(g) Length of free length of hose legs.
(h) Level of preservation-packaging, and packing (see 5.1)
(i) Vacuum application.
(j) Internal support when hose assembly is in vacuum service.

6.3 With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-24136 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 and information pertaining to qualification of products may be obtained from that activity. Appplication for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

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6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Supersession data. Part of this specification supersedes that portion of MIL-H-24135 for class IV hose and end fittings. Table II provides the appropriate cross references.

TABLE II. Supersession cross reference.

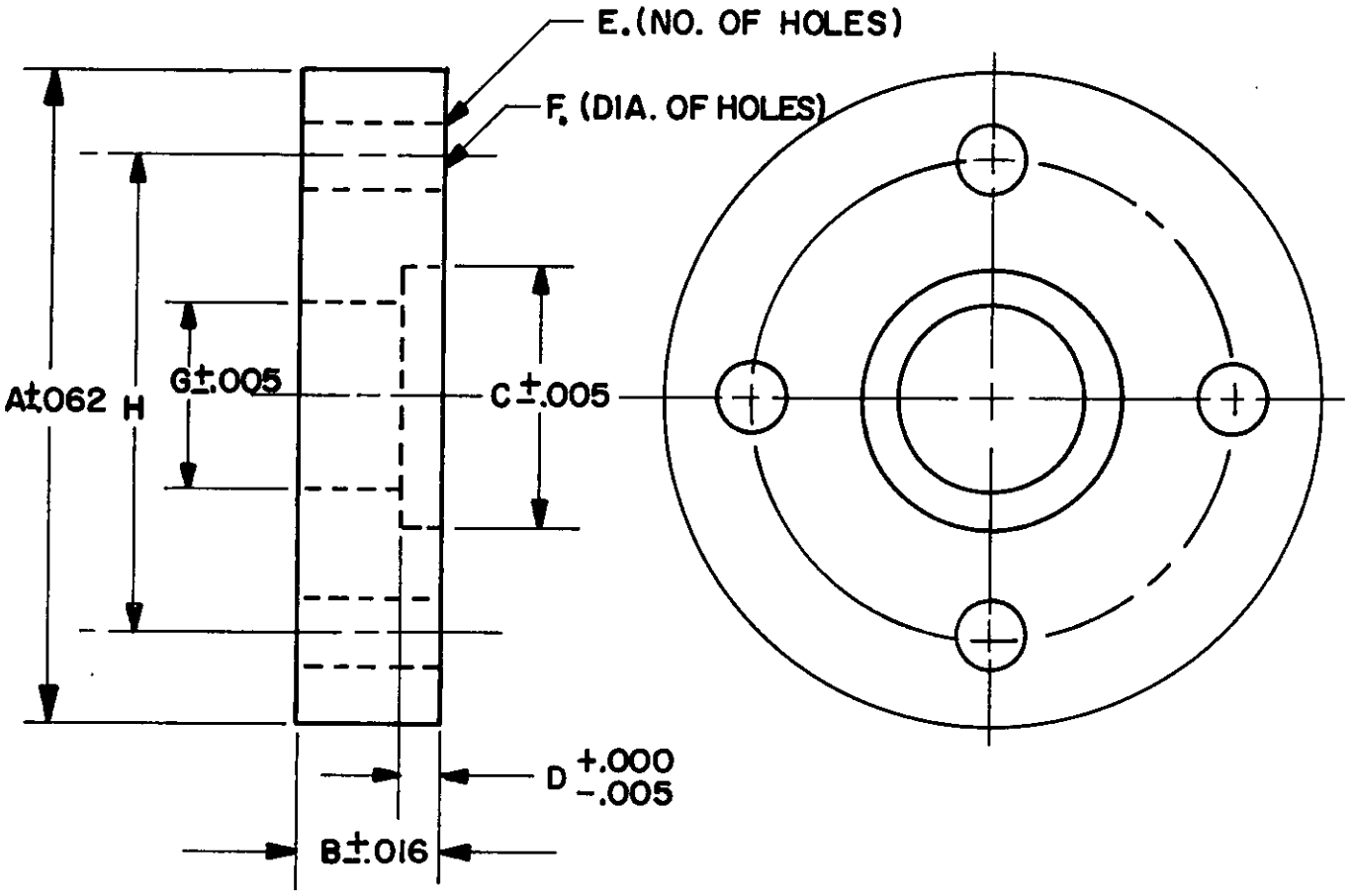
Specification/sheet	Old specification
MIL-H-24136/1	None
MIL-H-24136/2	MIL-H-24135, class IV
MIL-H-24136/3	None
MIL-H-24136/4	MIL-H-24136

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4730-N874)

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SH 12281

NOTES: 1. Dimensions are in inches.
2. See the following tables for dimensions:

Type I - table III - 400 lb/in^2 flange to MIL-F-20042
Type II - table IV - 150 lb/in^2 flange to MIL-F-20042 (with 400 lb/in^2 flange thickness)
Type III - table V - 250 lb/in^2 flange to MIL-F-20042
Type IV - table VI - 150 lb/in^2 flange to MIL-F-20042
Type V - table VII - 150 lb/in^2 flange to ASTM B16.5

3. Flanges in accordance with table IV are for replacement purposes only and are not to be used in new construction.

FIGURE 1. Flanges types.

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TABLE III. 400 lb/in^2 flange (MIL-F-20042).

Hose dash size	-16			-20	-24	-32			-40	
NPS size	1/2	3/4	1	3/4	1	1-1/4	1-1/2	2	2	2-1/2
A nominal	4.000	4.312	5.062	4.312	5.062	5.375	5.938	6.500	6.500	7.562
B minimum	0.687	0.687	0.750	0.687	0.750	0.812	0.812	0.812	0.812	0.937
C ± 0.005	1.515	1.515	1.515	1.859	2.140	2.765	2.765	2.765	3.296	3.296
D + 0.000 - 0.005	0.235	0.235	0.235	0.235	0.270	0.270	0.270	0.270	0.270	0.270
E	4	4	5	4	5	5	6	7	7	8
F nominal	0.562	0.562	0.688	0.562	0.688	0.688	0.688	0.688	0.688	0.812
G minimum	1.375	1.375	1.375	1.688	1.938	2.562	2.562	2.562	3.062	3.062
H nominal	2.875	3.188	3.750	3.188	3.750	4.062	4.625	5.188	5.188	6.000

Hose dash size	-48		-64		-80	-96	-128	-160
NPS size	2-1/2	3	3-1/2	4	5	6	8	10
A nominal	7.562	8.125	8.688	9.250	10.375	11.938	14.750	17.000
B minimum	0.937	0.937	1.000	1.000	1.062	1.187	1.312	1.437
C ± 0.005	3.796	4.515	5.245	5.295	6.393	7.393	9.870	12.080
D + 0.000 - 0.005	0.290	0.290	0.290	0.290	0.535	0.535	0.613	0.613
E	8	8	9	9	11	12	13	15
F nominal	0.812	0.812	0.812	0.812	0.812	0.937	1.062	1.06
G minimum	3.562	4.187	4.805	4.805	6.065	7.065	9.590	11.440
H nominal	6.000	6.562	7.125	7.688	8.812	10.188	12.750	15.000

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TABLE IV. 150 lb/in^2 flange (with 400 lb/in^2 flange thickness).

Hose dash size	-16			-20	-24	-32			-40	
NPS size	1/2	3/4	1	3/4	1	1-1/4	1-1/2	2	2	2-1/2
A nominal	3.562	3.812	4.250	3.812	4.250	4.500	5.062	5.562	5.562	6.125
B minimum	0.687	0.687	0.750	0.687	0.750	0.812	0.812	0.812	0.812	0.937
C ± 0.005	1.515	1.515	1.515	1.859	2.140	2.765	2.765	2.765	3.296	3.296
D + 0.000 - 0.005	0.235	0.235	0.235	0.270	0.270	0.270	0.270	0.270	0.270	0.270
E	3	4	4	4	4	4	6	6	6	6
F nominal	0.562	0.562	0.562	0.562	0.562	0.562	0.562	0.562	0.562	0.562
G minimum	1.375	1.375	1.375	1.688	1.938	2.562	2.562	2.562	3.062	3.062
H nominal	2.437	2.688	3.125	2.688	3.125	3.375	3.937	4.438	4.438	5.000

Hose dash size	-48		-64		-80	-96	-128	-160	-192
NPS size	2-1/2	3	3-1/2	4	5	6	8	10	12
A nominal	6.125	6.625	7.188	7.688	9.062	10.125	12.375	15.000	17.625
B minimum	0.937	0.937	1.000	1.000	1.062	1.187	1.312	1.437	1.500
C ± 0.005	3.796	4.515	5.245	5.295	6.393	7.393	9.870	12.080	14.535
D + 0.000 - 0.005	0.290	0.290	0.290	0.290	0.535	0.535	0.613	0.613	0.616 0.606
E	6	8	8	8	10	12	14	15	18
F nominal	0.562	0.562	0.562	0.562	0.688	0.688	0.688	0.812	0.812
G minimum	3.562	4.187	4.805	4.805	6.065	7.065	9.590	11.440	13.800
H nominal	5.000	5.500	6.062	6.562	7.812	8.875	11.062	13.438	16.062

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TABLE V. 250 lb/in^2 flange (MIL-F-20042).

Hose dash size	-16			-20	-24	-32			-40	
NPS size	1/2	3/4	1	3/4	1	1-1/4	1-1/2	2	2	2-1/2
A nominal	3.562	3.812	4.250	3.812	4.250	4.500	5.062	5.562	5.562	6.125
B minimum	0.687	0.687	0.750	0.687	0.750	0.812	0.812	0.812	0.812	0.937
C ± 0.005	1.515	1.515	1.515	1.859	2.140	2.765	2.765	2.765	3.296	3.296
D + 0.000 - 0.005	0.235	0.235	0.235	0.270	0.270	0.270	0.270	0.270	0.270	0.270
E	3	4	4	4	4	4	6	6	6	6
F nominal	0.562	0.562	0.562	0.562	0.562	0.562	0.562	0.688	0.688	0.688
G minimum	1.375	1.375	1.375	1.688	1.938	2.562	2.562	2.562	3.062	3.062
H nominal	2.437	2.688	3.125	2.688	3.125	3.375	3.937	4.438	4.438	5.000

Hose dash size	-48		-64		-80	-96	-128	-160	-192
NPS size	2-1/2	3	3-1/2	4	5	6	8	10	12
A nominal	6.125	6.625	7.188	7.688	9.062	10.125	12.375	15.000	17.625
B minimum	0.937	0.937	1.000	1.000	1.062	1.187	1.312	1.437	1.500
C ± 0.005	3.796	4.515	5.245	5.295	6.393	7.393	9.870	12.080	14.535
D + 0.000 - 0.005	0.290	0.290	0.290	0.290	0.535	0.535	0.613	0.613	0.616 0.606
E	6	8	8	8	10	12	14	15	18
F nominal	0.688	0.688	0.688	0.688	0.688	0.688	0.688	0.812	0.812
G minimum	3.562	4.187	4.805	4.805	6.065	7.065	9.590	11.440	13.800
H nominal	5.000	5.500	6.062	6.562	7.812	8.875	11.062	13.438	16.062

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TABLE VI. 150 lb/in^2 flange (MIL-F-20042).

Hose dash size	-16			-20	-24	-32			-40	
NPS size	1/2	3/4	1	3/4	1	1-1/4	1-1/2	2	2	2-1/2
A nominal	3.562	3.812	4.250	3.812	4.250	4.500	5.062	5.562	5.562	6.125
B minimum	0.375	0.438	0.438	0.438	0.438	0.438	0.438	0.438	0.438	0.500
C ± 0.005	1.515	1.515	1.515	1.859	2.140	2.765	2.765	2.765	3.296	3.296
D + 0.000 - 0.005	0.235	0.235	0.235	0.235	0.270	0.270	0.270	0.270	0.270	0.270
E	3	4	4	4	4	4	6	6	6	6
F nominal	0.562	0.562	0.562	0.562	0.562	0.562	0.562	0.562	0.562	0.562
G minimum	1.375	1.375	1.375	1.688	1.938	2.562	2.562	2.562	3.062	3.062
H nominal	2.437	2.688	3.125	2.688	3.125	3.375	3.940	4.438	4.438	5.000

Hose dash size	-48		-64		-80	-96	-128	-160	-192
NPS size	2-1/2	3	3-1/2	4	5	6	8	10	12
A nominal	6.125	6.625	7.188	7.688	9.062	10.125	12.375	15.000	17.625
B minimum	0.500	0.500	0.500	0.500	0.562	0.562	0.625	0.687	0.750
C ± 0.005	3.796	4.515	5.245	5.295	6.393	7.393	9.870	12.080	14.535
D + 0.000 - 0.005	0.290	0.290	0.290	0.290	0.535	0.535	0.613	0.613	0.616 0.606
E	6	8	8	8	10	12	14	15	18
F nominal	0.562	0.562	0.562	0.562	0.688	0.688	0.688	0.812	0.812
G minimum	3.562	4.187	4.805	4.805	6.065	7.065	9.590	11.440	13.800
H nominal	5.000	5.500	6.062	6.562	7.812	8.875	11.062	13.438	16.062

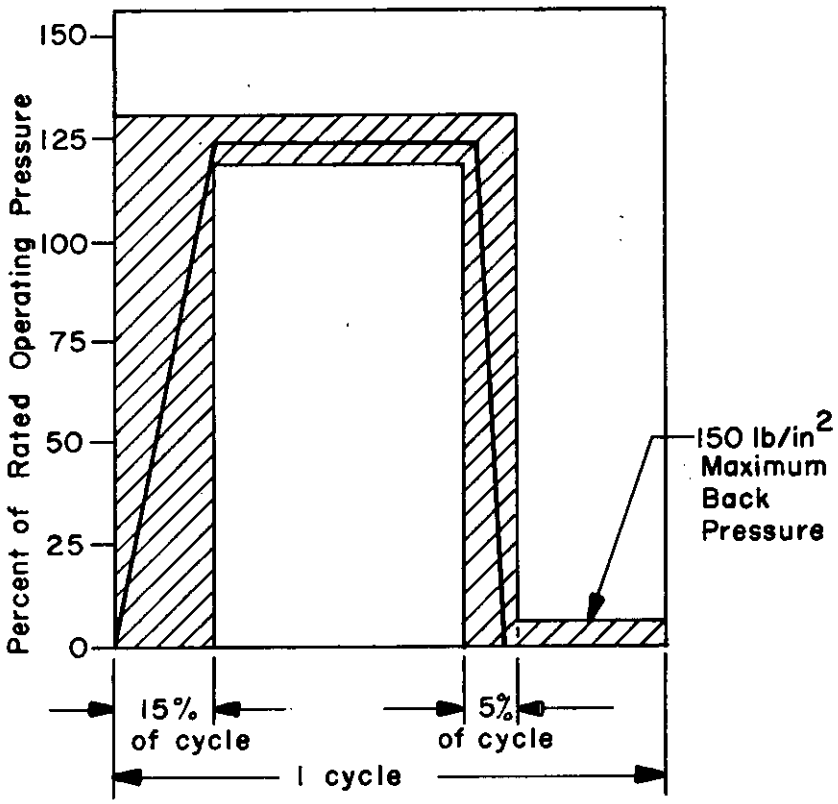
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TABLE VII. 150 lb/in^2 flange (ANSI B16.5).

Hose dash size	-16			-20	-24	-32			-40	
NPS size	1/2	3/4	1	3/4	1	1-1/4	1-1/2	2	2	2-1/2
A nominal	3.500	3.875	4.250	3.875	4.250	4.625	5.000	6.000	6.000	7.000
B minimum	0.359	0.422	0.422	0.422	0.422	0.422	0.422	0.422	0.422	0.482
C ± 0.005	1.515	1.515	1.515	1.859	2.140	2.765	2.765	2.765	3.296	3.296
D + 0.000 - 0.005	0.235	0.235	0.235	0.235	0.270	0.270	0.270	0.270	0.270	0.270
E	4	4	4	4	4	4	4	4	4	4
F nominal	0.625	0.625	0.625	0.625	0.625	0.625	0.625	0.750	0.750	0.750
G minimum	1.375	1.375	1.375	1.688	1.938	2.562	2.562	2.562	3.062	3.062
H nominal	2.375	2.750	3.125	2.750	3.125	3.500	3.875	4.750	4.750	5.500

Hose dash size	-48		-64		-80	-96	-128	-160	-192
NPS size	2-1/2	3	3-1/2	4	5	6	8	10	12
A nominal	7.000	7.500	8.500	9.000	10.000	11.000	13.500	16.000	19.000
B minimum	0.484	0.484	0.484	0.484	1.046	1.172	1.296	1.422	1.484
C ± 0.005	3.796	4.515	5.245	5.295	6.393	7.393	9.870	12.080	14.535
D +0.000 -0.005	0.290	0.290	0.290	0.290	0.535	0.535	0.613	0.613	0.616 0.606
E	4	4	8	8	8	8	8	12	12
F nominal	0.750	0.750	0.750	0.750	0.875	0.875	0.875	1.000	1.000
G minimum	3.562	4.187	4.805	4.805	6.065	7.065	9.590	11.440	13.800
H nominal	5.500	6.000	7.000	7.500	8.500	9.500	11.750	14.250	17.000

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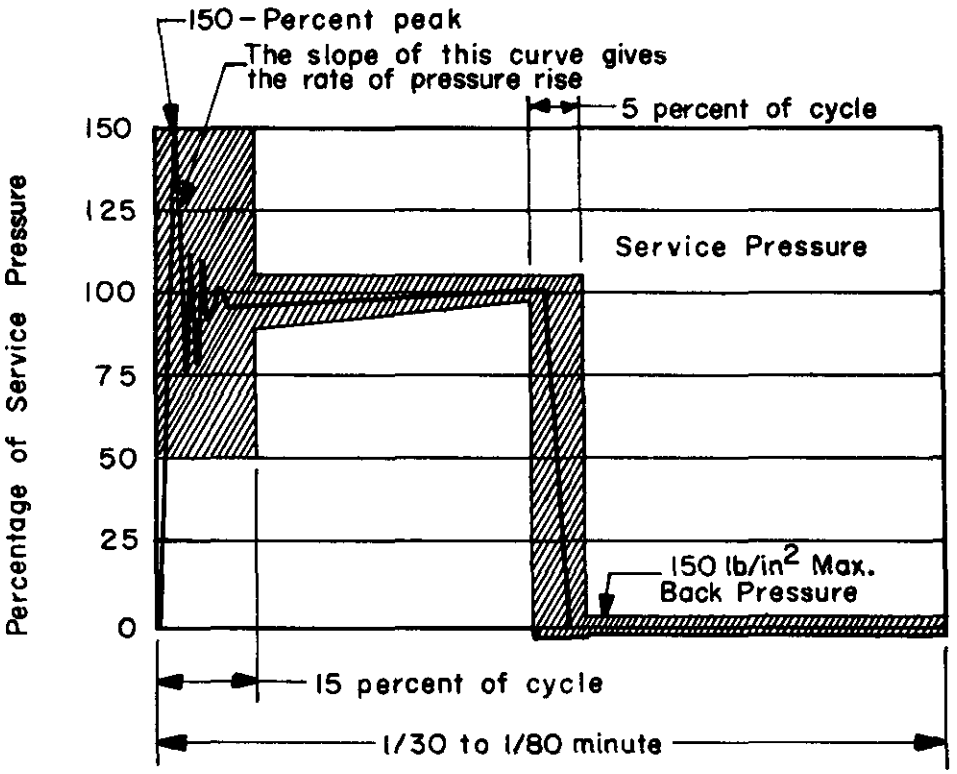
SH 8053A

For hose sizes larger than 2 inches (-32)

NOTE:
The curve is the approximate pressure time cycle for the pressure impulse to be used in performing the impulse test specified in 4.6.5. The actual pressure time cycle obtained when performing this test, should fall within the shaded area.

FIGURE 2. Pressure time curve for impulse test.

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SH 12652

For hose sizes up to 2 inches (-32)

The curve shown on the figure is the approximate pressure time cycle for the pressure impulses to be used in performing the hydraulic impulse test specified in 4.6.5. The actual pressure time obtained, when performing this test, should fall within the shaded area.

FIGURE 3. Pressure time curve for hydraulic impulse test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-24136A(SH)	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, And End Fittings, Reusable, For Flexible Hose Assemblies

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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